

Safe Mine Vehicle Operation in Fog & Low Visibility



NMDC Limited is India's largest iron ore producer, producing approximately **53 Million Tonnes Per Annum** across three fully mechanized mining complexes — BIOM-Kirandul, BIOM-Bacheli (Chhattisgarh) and Donimalai (Karnataka).

Bailadila alone contributes nearly 37 MTPA — the single largest driver of NMDC's output. Under the National Steel Policy, NMDC targets **100 MT by 2030**, with roughly 80 MT expected from Bailadila. It is the region that has to carry most of NMDC's future growth.

THE PROBLEM

The Monsoon Challenge

Bailadila's growth target runs headlong into its geography and climate. From June to October, heavy rainfall, strong winds, high humidity and thick fog are routine — and on hilltop haul roads, visibility can collapse to almost nothing.

3–5 m

Visibility in dense fog

Jun–Oct

Monsoon season

~5 months

Every single year

At 3–5 m visibility, Heavy Earth Moving Machinery, particularly ore-hauling dumpers can no longer move safely at normal speed. For nearly five months of the year, the very terrain that makes Bailadila productive also makes it one of the hardest places in the country to run mining operations safely.

WHY IT MATTERS

The Stakes and the Shortfall

THE URGENCY

Dense fog during the monsoon creates major operational and safety challenges in the Bailadila mines. Poor visibility restricts dumper movement, forcing operators to slow down or halt entirely.

This directly drives up haul-cycle times, cuts fleet productivity, and reduces ore evacuation, at the exact time NMDC needs Bailadila scaling toward 80 MT by 2030.

Beyond the productivity hit, the risk of collisions and operational disruption rises sharply every monsoon. This isn't occasional as it recurs for nearly five months every year, so any fix must run reliably through the entire monsoon window.

GAPS IN CURRENT SOLUTIONS

Existing visibility aids mostly rely on a single sensing method which is typically vision-based and is exactly what fails first as fog thickens.

Current systems are reactive, not predictive: operators slow down only once a hazard is already close, instead of the system anticipating risk from relative motion and stopping distance.

There's no shared awareness between vehicles, and safety margins don't adjust dynamically as sensor confidence degrades in fog, noise or rain thus leaving a system that depends almost entirely on manual caution.

PROPOSED IDEA

Our Solution: First Light

First Light is a multi-sensor, predictive safety system. Instead of depending on any single sensor, it fuses LiDAR, camera, radar, GNSS, IMU and wheel-state data into one continuous picture of the vehicle's surroundings, position and motion.



Predictive hazard tracking

Tracks relative motion, estimates time-to-collision and checks safe stopping distance thus shifting the vehicle from cruise to caution to full stop as needed.



V2V hazard & fog broadcast

Vehicles share hazard and fog observations over V2V comms, accounting for real-world message delay and loss.



Automated junction safety

Identifies conflicts between approaching vehicles and assigns yielding which helps in preventing collisions without operator intervention.



6 sensor modalities fused

24/7 predictive monitoring

TECHNICAL APPROACH

3-Layer Fog-Safe Vehicle System

A three-layer integrated system ensuring safe, fog-resilient vehicle operations across open-cut mining environments.



Layer 1 — On-Vehicle Sensing

Radar array · Thermal camera · DGPS positioning



Layer 2 — V2V / V2I Mesh

V2V comms · Infrastructure broadcast · Delay & loss aware



Layer 3 — Command & Control

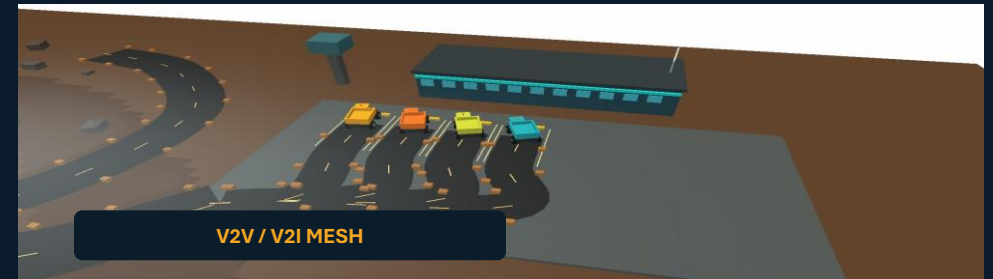
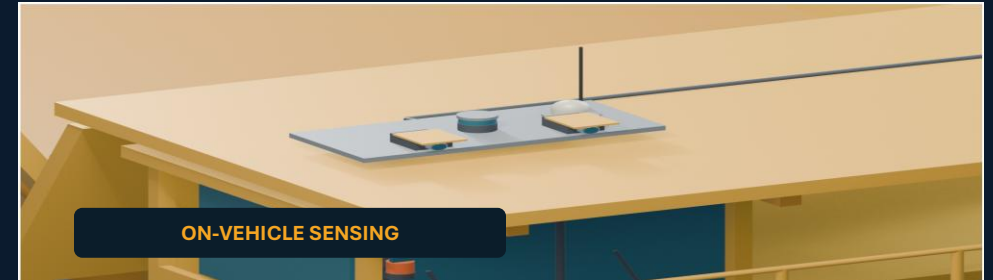
Control room dashboard · Digital twin · Site-wide oversight

Real-time
fog detection

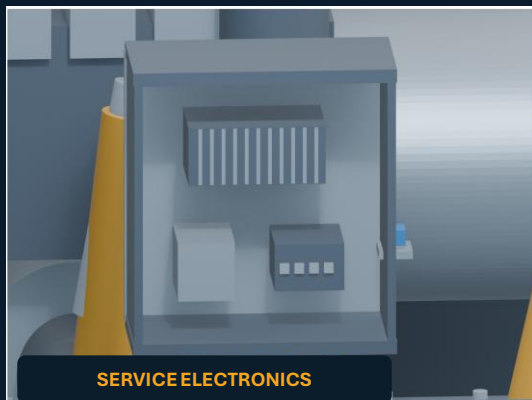
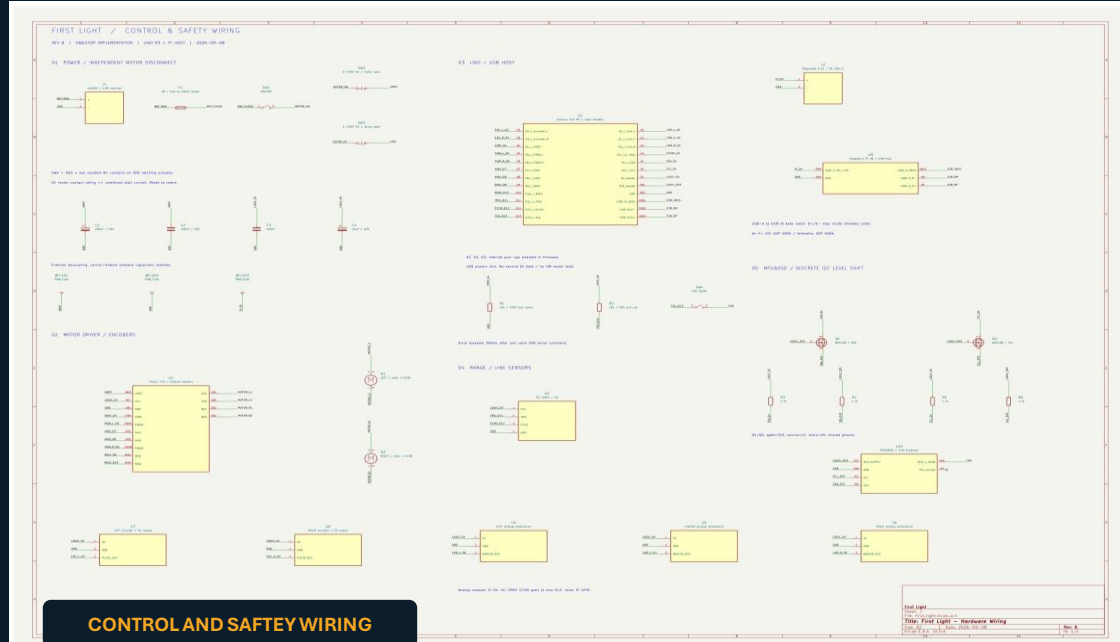
Sub-meter
positioning

Mesh < 50 ms

Full digital twin



Built Lean. Engineered Safe.



PROTOTYPE COMPONENTS

Arduino Uno

TB6612FNG motor driver

HC-SR04 ultrasonic

MPU6050 IMU

Wheel encoders

Raspberry Pi (telemetry / V2V)

Low-cost, readily available parts validate the full decision and control logic. The architecture is modular, the same pipeline scales to industrial LiDAR / radar without a redesign, letting NMDC validate cheaply before fleet-wide investment.

BUILT FOR RELIABILITY

- Multi-sensor fusion stays reliable even when individual sensors lose confidence in fog.
- Fail-safes (NC e-stop, watchdog, command-timeout) default the vehicle to a safe stop on any fault.
- Predictive, not reactive and calculates risk before a hazard becomes critical.
- Tested against fog, sensor noise, comms delay & message loss, not just ideal cases.

SIH 2026 – SIH26007

Our Team

Team First Light

Nitish Kumar

Team Leader

Akul Batra

Team Member

Kritika Purbey

Team Member

Tejas Gupta

Team Member

Ghanansh Kuradia

Team Member

Anika Mittal

Team Member

